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Removal of a textile dye by adsorption on synthetic calcium phosphates

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ABSTRACT

The objective of this study was to investigate the ability of synthetic calcium phosphates to remove a textile dye from aqueous solutions. The adsorption of Reactive Yellow 4 (RY4) on apatitic tricalcium phosphate (PTCa) and apatitic octocalcium phosphate (OCPa) from aqueous solutions was investigated. The kinetic study shows that equilibrium is quickly obtained for both phosphates. The kinetic analysis showed that the pseudo-second-order model had the best fit to the experimental data. The adsorption isotherms of the dye by both phosphates were correlated with Langmuir and Freundlich models depending on the phosphate studied. In addition, the maximum adsorption capacity decreased from 35.50 to 31.86 mg/g and 29 to 19.85 mg/g for PTCa and OCPa, respectively, when the temperature was increased from 25 to 45 °C. The thermodynamic evaluation of RY4 adsorption on both phosphates revealed that the adsorption phenomenon was a spontaneous, exothermic and physical process. Accordingly; synthetic calcium phosphates have been shown to be a very efficient and low-cost adsorbent, and a promising alternative for eliminating dyes from industrial wastewaters.

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Contents

1. Introduction . . . 10
2. Materials and methods . . . 11
   2.1. Materials used . . . 11
      2.1.1. Adsorbents . . . 11
   2.2. Adsorbate . . . 11
   2.3. Experimental protocol . . . 11
3. Results and discussion . . . 12
   3.1. Effect of solid quantity . . . 12
   3.2. Kinetics of adsorption . . . 13
   3.3. Study of adsorption . . . 14
   3.4. Rate of decolorization . . . 14
   3.5. Effect of temperature on adsorption equilibrium . . . 15
4. Conclusion . . . 16
References . . . 16

1. Introduction

The problems of water resources and pollution are increasingly on the agenda. Many industries like textile or cosmetic industry use for their end products of organic compounds in particular colors. In Morocco the textile industry, representing 31% of all Moroccan industries whose reactive dyes are widely used for dyeing wool and nylon. In the textile industry, about 1000 l of water are used per 1000 kg of clothes processed in dyeing [1]. Thus the releases of this industry are loaded dyes [2] and present a real danger to the environment due to the high water consumption and the important wastewater discharges rejected without any treatment. The first step of removal of dyes in the wastewater is a physico-chemical treatment by adding chemical reagents; then a biological step occurs using activated sludge in aeration [3]. Note, however, that dyes are poorly biodegradable where there is a limiting factor of biological treatment.

Other physico-chemical treatments are currently used. They include the adsorption on clay as sepiolite, zeolite, montmorillonite, smectite, bentonite, alunite and perlite [4–7]. All these methods are significantly different in terms of discoloration, process and economic

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balance [8,9]. Recently, calcium phosphates are very much studied in the removal of heavy metal ions [10], fluoride [11], nitrobenzene [12] and protein [13]. Barka et al. [14,15] have reported that the natural phosphate and poorly synthesized crystalline hydroxyapatite were used for the removal of dyes from aqueous solutions. The results showed that both materials were effective to remove dyes.

Hydroxyapatite (Ca10(PO4)6(OH)2, HAP), a bioceramic analogous to the mineral component of bone, has been used extensively in such fields as bone repairs, bone implant and bioactive materials due to its excellent biocompatibility, slow biodegradation, osteoconductivity and mechanical properties [16–19]. Calcium phosphates, especially apatites, present remarkable interest, their availability structure, ionic exchange property, adsorption affinity, and their characteristic to establish bonds with organic molecules of different sizes, have conferred to this material to attract more attention. In addition, this material can be efficient matrixes of water purification.

In this context, we are interested in the possibility of using synthetic calcium phosphate as apatitic tricalcium phosphate, which is a solid with atomic ratio of Ca/P, equal to 1.5, and octocalcium phosphate apatite of Ca/P ratio, equal to 1.33, in order to eliminate the reactive dye; Reactive Yellow 4 (RY4), from aqueous solution. The Langmuir and Freundlich models were used to fit the equilibrium isotherm.

2. Materials and methods

2.1. Materials used

2.1.1. Adsorbents

The adsorption tests were conducted on two calcium phosphates: apatitic tricalcium phosphate (PTCa) and phosphate apatitic octocalcium (OCPa).

The first one (PTCa) was prepared at room temperature by a double decomposition method [20]. The solution A (47 g of calcium nitrate Ca $(NO_3)_2 \cdot 4H_20$ (Scharlau, Spain) in 550 ml of distilled water + 20 ml of ammonia solution) was added quickly at room temperature into the solution B (26 g of di-ammoniumhydrogenphosphate$(NH_4)_2HPO_4$ (Riedel-de Haën, Germany) in 1300 ml of distilled water + 20 ml ammonia solution). The precipitate was filtered, washed, and dried at 80 °C for 24 h.

The second one, OCPa was also developed by a fast coprecipitation in a rural hydroalcoholic solution (50% ethanol by volume) between a calcium solution [A'] and a basic phosphate solution [B'] [21]. The solution A'(30 mmol of nitrate calcium Ca $(NO_3)_2 \cdot 4H_20$ (Scharlau, Spain) in 100 ml of distilled water and 100 ml of ethanol) was immediately poured at 37 °C into a solution B'(30 mmol of di-ammoniumhydrogenphosphate$(NH_4)_2HPO_4$ (Riedel-de Haën, Germany) in 250 ml of distilled water, 45 ml of a solution of ammonia and 295 ml of ethanol). The coprecipitate was filtered and then dried in an oven at 80 °C for 24 h.

Both phosphates were characterized by chemical and physical analysis. The calcium content in the solid was determined by complexometry with EDTA and the phosphate ion content by spectrophotometry of phospho-vanado-molybdic acid. The specific surface area was determined according to the BET method using N_2 adsorption (Table 1). Infrared spectroscopy IR was carried out after dispersion of anhydrous KBr (about 2 mg product to 200 mg of KBr) using TENSOR 27 spectrophotometer (Bruker Optics, Germany) (Figs. 1 and 2).

Table 1
Characteristics of phosphates used.

Name	Ca/P	Specific surface (m^2/g)
PTCa	1.50	62
OCPa	1.33	58

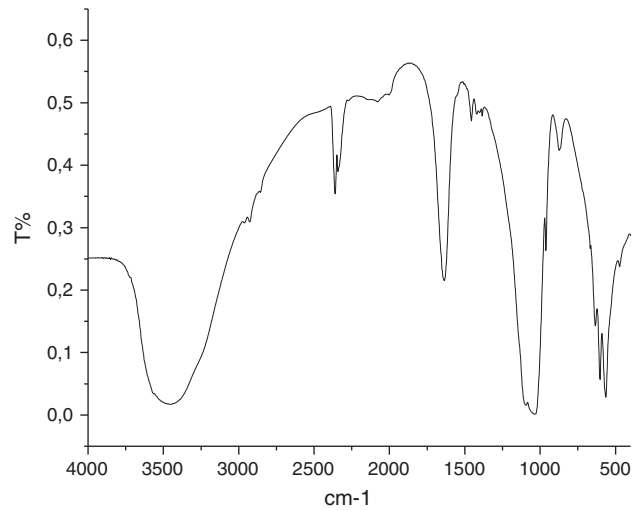


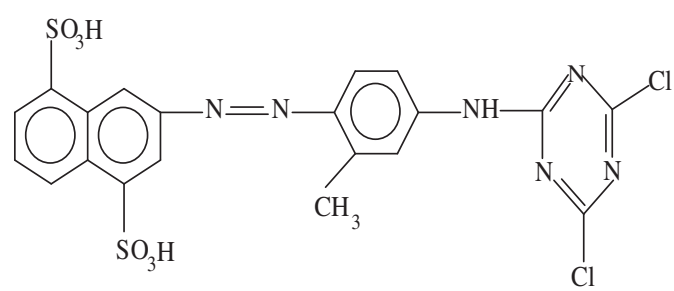
Fig. 1. Infrared absorption spectrum of PTCa of Ca/P = 1.5.

These spectra correspond with the two phosphate apatitic compounds

Ca9□(PO_4)5(HPO_4)(OH)□	PTCa
Ca8□2(PO4)3.5(HPO4)2, 5(OH)0, 5□1.5	OCPa.

2.2. Adsorbate

The Reactive Yellow 4 (C.I) was obtained from a textile firm as a commercially available dye formulation and was used without further purification. It is a soluble dye in water due to the presence of two solubilizing groups (SO_3H).



Chemical structure of dye (C.I. Reactive Yellow 4).

The yellow color is due to the grouping diphenylparaazolonique. The grouping dichlorotriazinique ensures reactivity of the molecule with the textile fiber. The solutions ware prepared by dissolving the required amount of dye in distilled water .The concentration of the dye was determined at 385 nm, using UV spectrophotometer ("UV-2005", Selecta, Spain).

2.3. Experimental protocol

The effect of solid quantity on removal of Reactive Yellow 4 (RY4) was investigated in batch experiments by adding various amounts of adsorbent in the range of 0.05–0.4 g powder into a test tube containing 10 ml of the dye solution. The initial dye concentrations of the solutions were fixed at 100 mg/l, for all batch experiments. The suspension was then stirred for 1 min, after which time the solution was coagulated and settled and the supernatant was analyzed for the remaining dye.

To study the kinetics of adsorption of the dye at 25 °C, a volume of 10 ml of the dye at concentration 100 mg/l was placed in contact with 200 mg of the adsorbent in a test tube. The mixture was stirred at constant speed (500 rpm) for one minute and then placed in a water

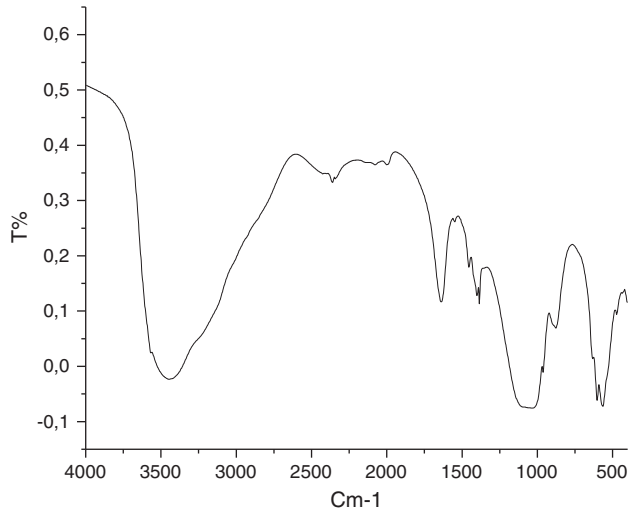


Fig. 2. Infrared absorption spectrum of OCPa of Ca/P = 1.33.

bath at 25 ° C. After, the solid was separated from the mother solution by filtration through a sintered glass and the dye concentration was determined using the UV–vis spectrophotometer.

The quantity of dye per which was fixed gram of adsorbent was given by the following equation:

$$Q_t = \frac{(C_0 - C_t)V}{m}$$

where:

Qt the quantity of dye in mg per gram of adsorbent
Co and Ct are respectively the initial concentration and time t of the dye (mg/l).
V volume of solution (l).
m mass of adsorbent used (g).

The classical models of Langmuir and Freundlich characterizing the formation of a monolayer are used for their simplicity of implementation.

Thus the linearized Langmuir equation [22] allowed us to determine the characteristic parameters of adsorption, namely the amount adsorbed at saturation Q∞ and the constant interaction adsorbate–adsorbent b.

$$\frac{1}{Q_{ads}} = \frac{1}{(bQ_\infty)C_{eq}} + \frac{1}{Q_\infty}$$

Freundlich's law is a purely empirical relationship [23]:

$Y = aX^m$ with $m \leq 1$.

Y the quantity of substance adsorbed per unit area (or mass) of adsorbent.
X concentration of adsorbate in solution at equilibrium.

The parameters a and m characterize the adsorbent–adsorbate pair. The linear transform of this equation is:

$$log(Y) = log(\alpha) + m\ log(X).$$

The variation of log (Y) versus log (X) allows to determine m and a. It should be noted that when the amount adsorbed is very low, it is generally the Freundlich's law that describes the mode of adsorption. For a high recovery rate, it is the law of Langmuir which allows describing this process.

3. Results and discussion

3.1. Effect of solid quantity

The results concerning the effect of the solid quantity are shown in Fig. 3. As indicated, 38 and 50% of RY4 were removed at the initial quantity of 0.05 g of PTCa and OCPa, respectively. The removal of dye increased with increasing solid quantity up to 0.2 g and reached to 89 and 61.86% for PTCa and OCPa, respectively, at this quantity. However it is observed that after a dosage of 0.2 g of PTCa, there was no significant change in the percentage removal of dye. This may be due to the overlapping of active sites at higher dosage. So, there was not any appreciable increase in the effective surface area resulting due to the conglomeration of particles. But the amount of dye retained after a dosage of 0.2 g of OCPa decreased, the reason for this phenomenon was may be that, for the dye was distributed on greater amount of surface and therefore resulted in the reduction of adsorption amount on unit mass adsorbent. So, 0.2 g was considered as the optimum dose of both phosphate powders and was used for further study.

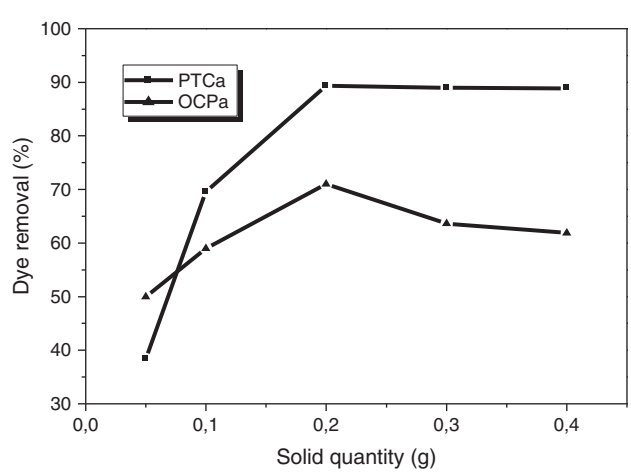


Fig. 3. Effect of initial quantity of removal of RY4 on phosphates used (initial dye concentration: 100 mg/l, pH: 5.69, stirring time: 1 min, T: 25 °C).

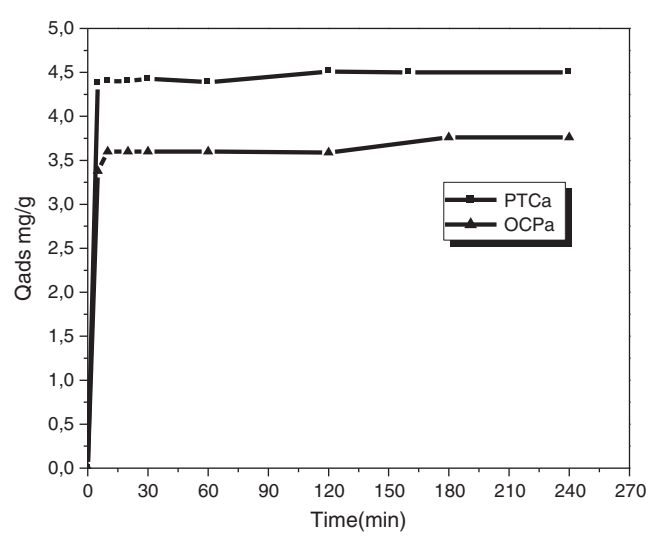


Fig. 4. Kinetics of adsorption of RY4 on phosphates used at 25 °C (solid quantity: 200 mg, initial dye concentration: 100 mg/l, pH: 5.69, stirring time: 1 min).

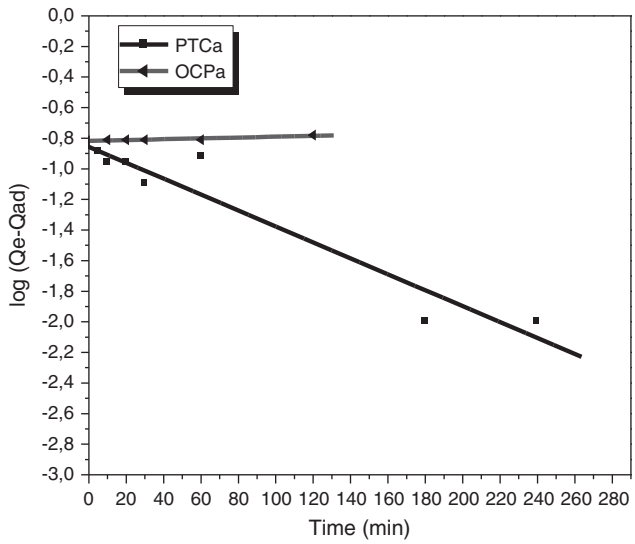


Fig. 5. Determination of rate constants of pseudo first order adsorption of Reactive Yellow 4 by phosphates.

3.2. Kinetics of adsorption

For both phosphates, the equilibrium was quickly reached. The maximum adsorption of the dye was observed in the first half hour of contact. This fast rate can be explained by the high number of active sites available at the beginning of relative adsorption sites remaining after some time.

Fig. 4 presents the kinetic results of the adsorption process. The evolution of the amount adsorbed as a function of time shows that the kinetics of adsorption of the dye used on both phosphates have the same rate. The equilibrium was reached within a few minutes, by establishing a well-formed plateau, indicating also that the phosphates are very effective adsorbents in a very short time for this textile dye.

From the model established by Lagergren [24], the rate constant of adsorption of the pseudo first order is derived. However, as Ho and Mc Kay [25,26], we opted for a kinetic model of order 2 to interpret the best possible reaction mechanism.

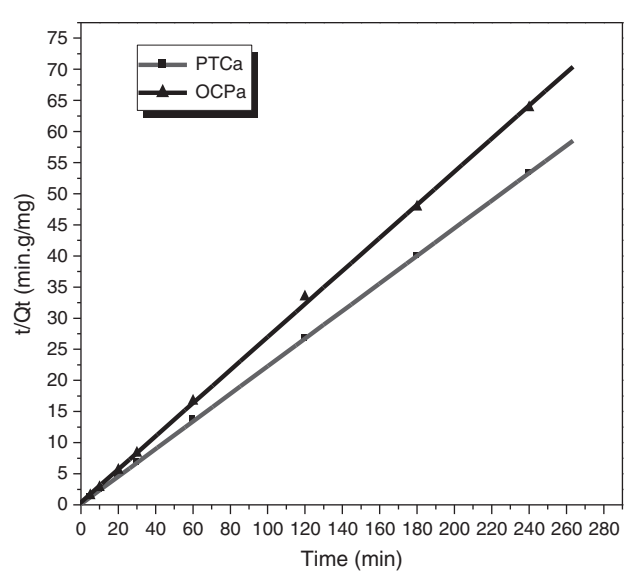


Fig. 6. Determination of rate constants of pseudo second order adsorption of Reactive Yellow 4 by phosphates. adsorption of Reactive Yellow 4 by phosphates.

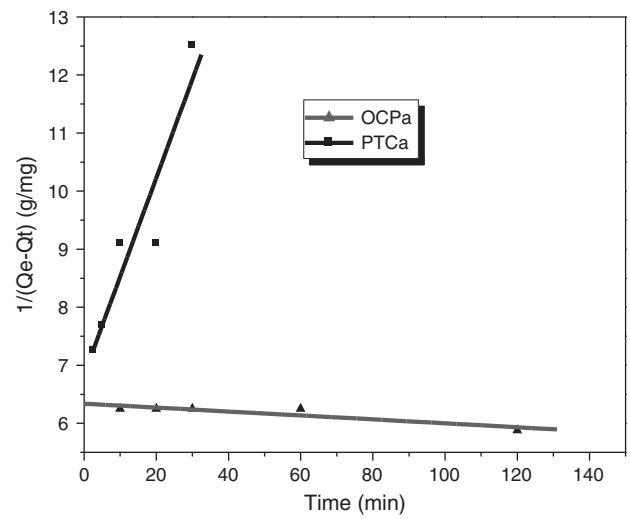


Fig. 7. Determination of rate constants of second order adsorption of Reactive Yellow 4 by phosphates.

For the pseudo first order rate constant K_1 is given by the following equation:

$$log(Q_e - Q_t) = log(Q_e) - \frac{K_1}{2,3}t.$$

For the pseudo second order rate constant K_2 is given by the following equation:

$$\frac{t}{Q_t} = \frac{1}{2K_2Q_e^2} + \frac{t}{Q_e}.$$

For the second order rate constant K_3 is given by the following equation:

$$\frac{1}{(Q_e - Q_t)} = \frac{1}{Q_e} + K_3t.$$

With:

Q_e Amount of adsorbate at equilibrium per gram of adsorbent (mg/g).
t contact time (min).

K_1, K_2 and K_3 are rate constants of adsorption, respectively, for the pseudo first order (min^{-1}), the pseudo second order (g min/mg) and the second order (min^{-1} g/mg).

The rate constants of adsorption of the dye on phosphate apatitic tricalcium and apatitic phosphate octocalcium for the pseudo first order, pseudo second order and second order were determined

Table 2
Constant rates of pseudo first order.

Adsorbents	Q_e (mg/g)	K_1 (min^{-1})	R^2
PTCa	0.139	0.012	0.9586
OCPa	0.152	0.116	0.9068

Table 3
Constant rates of pseudo second order.

Adsorbents	Q_e (mg/g)	K_2 (min^{-1}g/mg)	R^2
PTCa	4.51	0.2192	0.99998
OCPa	3.76	0.0868	0.99976

Table 4
Constant rates of the second order.

Adsorbents	Q_e (mg/g)	K_3 (min^{-1}g/mg)	R^2
PTCa	0.1464	0.16986	0.91973
OCPa	0.1577	−0.00338	0.90683

Table 5
The amount of dye adsorbed at equilibrium by the PTCa and the OCPa.

Adsorbents	Q_e exp (mg/g)	Q_e cal (mg/g) pseudo first order	Q_e cal (mg/g) pseudo second order	Q_e cal (mg/g) second order
PTCa	4. 50	0.139	4.51	0.1464
OCPa	3.76	0.152	3.76	0.1577

graphically (Figs. 5, 6 and 7). For each pair Dye-PTCa or Dye-OCPa, these constants were evaluated.

The determination of the different rate constants (Tables 2–4) shows that the model of pseudo second order with a good correlation coefficient ($R^2 = 0.999$) is the most reliable.

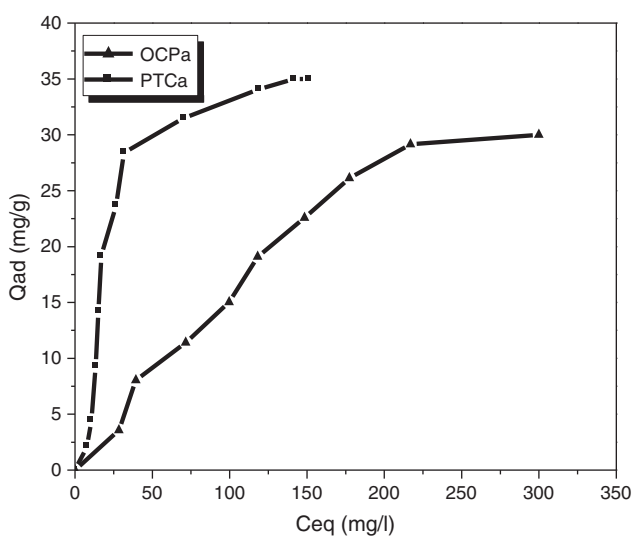


Fig. 8. Adsorption isotherms of Reactive Yellow 4 by phosphates. (Solid quantity: 200 mg, stirring time: 1 min, T: 25 ° C).

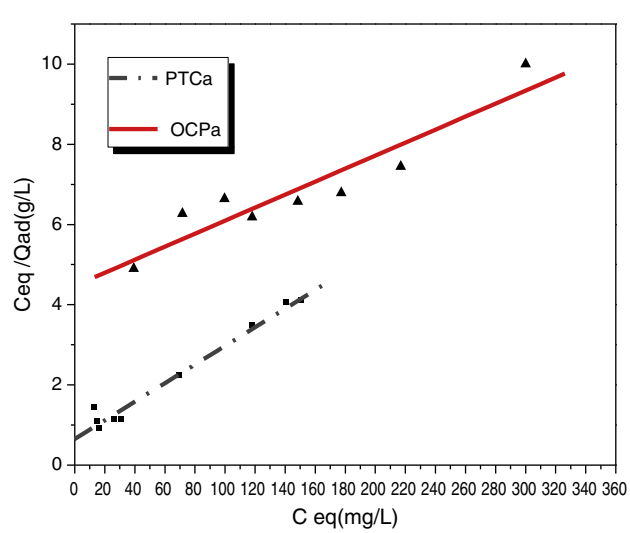


Fig. 9. Modeling of adsorption isotherm Reactive Yellow 4 from phosphates by Langmuir model.

Table 6
Values of the constants of Langmuir model.

Adsorbants	Q∞ (mg/g)	b (l/g)	R^2
PTCa	40.85	47.6	0.987
OCPa	61.65	3.63	0.933

Meanwhile, the comparison of calculated values of Q_e with experimental values (Table 5) confirms that the adsorption kinetics are pseudo second order. This observation is in agreement with many other studies of kinetics of adsorption of dyes [27,28].

3.3. Study of adsorption

The adsorption isotherms of RY4 on the two phosphates was determined by using 200 mg of adsorbent in contact for 6 h with 10 ml of different solutions with a concentration of dye ranging from 100 to 900 mg/l.

The evolution of the amount adsorbed as a function of its equilibrium concentration in the medium is shown in Fig. 8.

It is observed that the amount of absorbed dye increases more quickly for low concentrations in solution and then reaches a plateau; that can be explained by a monolayer adsorption type.

The results of modeling of adsorption isotherms by the Langmuir model are shown in Fig. 9.

According to the values of correlation coefficients (Table 6), one can say that the Langmuir model appears appropriate only for modeling the adsorption isotherms on apatitic tricalcium phosphate.

According to the Freundlich model, the results of modeling of adsorption isotherms of the dye by the apatitic tricalcium phosphate and phosphate octocalcium apatite are shown in Fig. 10.

According to the values of the correlation coefficients (Table 7), one observes that the Freundlich model is not valid to describe the isotherms of adsorption on apatitic tricalcium phosphate in the whole domain of concentrations studied.

3.4. Rate of decolorization

The percentage of decolorization was calculated from the relationship:

$$\%\ \text{of decolorization} = \frac{(C_0 - C_t)}{C_0}.$$

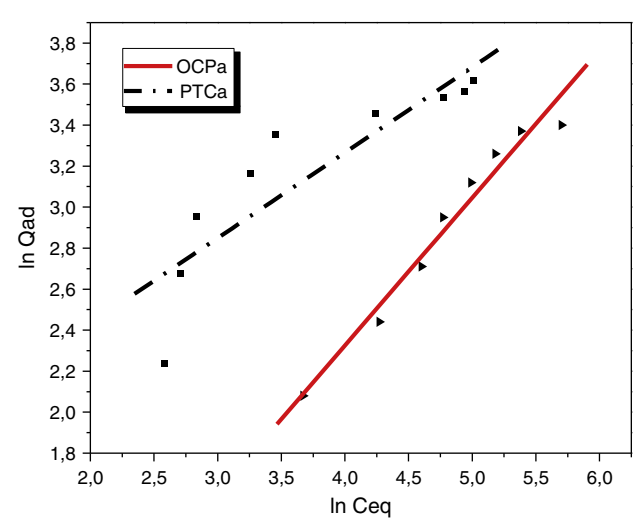


Fig. 10. Modeling of adsorption isotherm Reactive Yellow 4 from phosphates by using the model of Freundlich.

Table 7
Values of the constants of Freundlich model.

Adsorbents	m	a (l/g)	R^2
PTCa	0.42	4.95	0.881
OCPa	0.72	0.57	0.983

Table 8
Percentage of dye removal from phosphates.

Time (min)	% of decolorization of RY4	
	PTCa	OCPa
0	0	0
5	87.61	67.68
10	87.93	72.00
20	87.93	72.00
30	88.54	72.00
60	87.74	72.00
120	89.98	71.84
180	89.82	75.20
240	89.82	75.20

C_o and C_t: are respectively the initial concentrations and in the time t of the dye (mg/l).

The elimination rates of RY4 are 90% for the apatitic tricalcium phosphate and 75% for phosphate octocalcium apatite (Table 8) (Fig. 11).

3.5. Effect of temperature on adsorption equilibrium

On Fig. 12, we represented the evolution of the amount adsorbed as a function of temperature.

There is an increase in temperature which causes a decrease in the adsorption capacity of dye at equilibrium regardless of the phosphate used. The decrease in the temperature range 25–45 °C, can be explained by the fact that adsorption is an exothermic process.

Thus, generally, the phenomenon of adsorption is accompanied by a heat exchange [29,30] which can be either exothermic or endothermic. The main criterion which leads to a chemisorption or physisorption is the determination of the enthalpy ΔH.

The thermodynamic parameters such as Gibbs free energy (ΔG), enthalpy (ΔH), and entropy (ΔS) were determined using the equations [31]:

$$\Delta G = -RTLnK_c$$

$$\Delta G = \Delta H - T\Delta S$$

$$LnK_c = \frac{\Delta S}{R} - \frac{\Delta H}{RT}$$

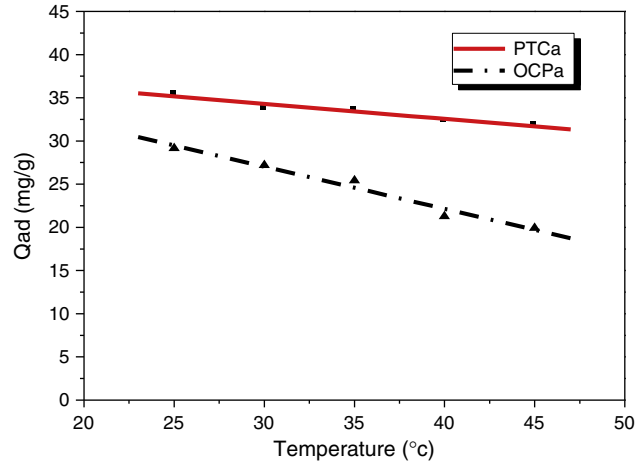


Fig. 12. Effect of temperature on the equilibrium adsorption of Reactive Yellow 4 on phosphates. (Solid quantity: 200 mg, initial dye concentration: 800 mg/l, pH: 6.86, stirring time: 1 min, contact time: 6 h).

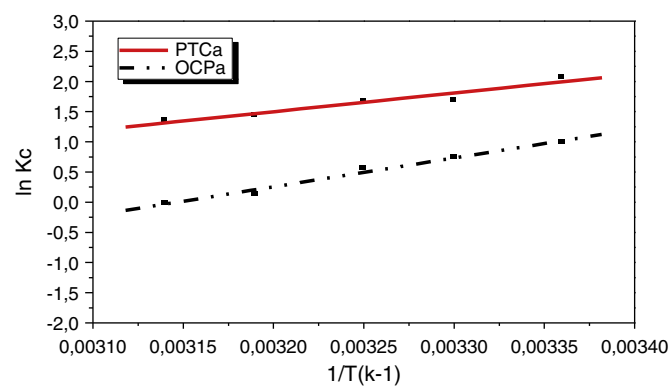


Fig. 13. Enthalpies and entropies' determination of adsorption of Reactive Yellow 4 on phosphates.

$$K_c = \frac{(C_0 - C_{eq})}{C_{eq}}$$

with:

KC	Equilibrium constant.
ΔG	Gibbs free energy (Joule/mole).
ΔH	Enthalpy (Joule/mole).
ΔS	Entropy (Joule/mol K).
T	Absolute temperature (K).
C_0	Initial concentration of the adsorbate.
C_{eq}	Equilibrium concentration.
R	Gas constant.

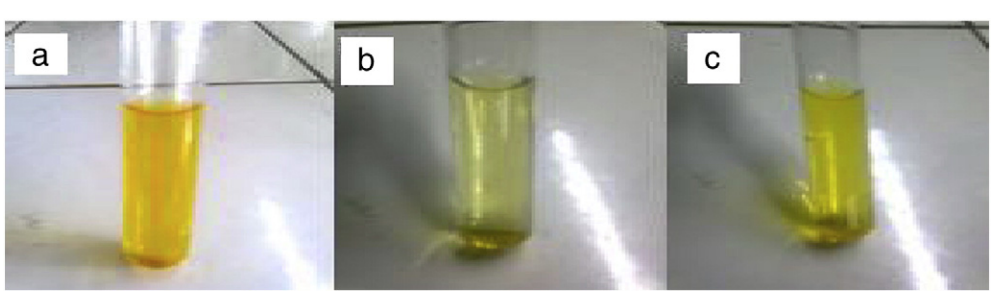


Fig. 11. Picture of untreated solution 100 mg/l of RY4 a) and supernatant after treatment with: PTCa b) and OCPa c).

Table 9
Thermodynamic parameters of adsorption of the dye from phosphates.

Adsorbents	ΔH (K Joule/mole)	ΔS (Joule/mole K)	R^2
PTCa	−25.75	−69.93	0.9601
OCPa	−39.67	−124.83	0.9914

Table 10
Values of Gibbs free energy of adsorption of the dye from phosphates.

Adsorbents	T (°C)	ΔG (K Joule/mole)	Kc
PTCa	25	−4.91	7.97
	30	−4.56	5.43
	35	−4.21	5.31
	40	−3.86	4.22
	45	−3.51	3.91
OCPa	25	−2.47	2.69
	30	−1.85	2.12
	35	−1.22	1.74
	40	−0.61	1.14
	45	0.02	0.99

The curve Ln Kc as a function of the inverse of temperature (1/T) (Fig. 13) allowed us to determine ΔH and ΔS for the two phosphates. The straight lines obtained show good correlations coefficients.

Negative values of ΔH confirm that the adsorption of dye on the various phosphates is an exothermic process (Table 9). Moreover, we can conclude that a physical adsorption in the values of ΔH are less than 40 KJoule/mole. However, the details of adsorption mechanism need to be further investigated.

The variations of negative entropy ΔS accompanying the adsorption of dye on the two phosphates show that we reached a more orderly system. They confirm the decrease in random molecules of the dye in the solid–liquid interface.

Changes in Gibbs free energy ΔG are negative (Table 10) which indicate that adsorption of the dye on the two phosphate is a spontaneous or natural process.

4. Conclusion

This study shows that the use of these two phosphates reduces the pollution discharge textiles in an important amount. In both cases, the dye is adsorbed quickly. The kinetics of adsorption are similar (pseudo second order) for both phosphates. The Langmuir model describes satisfactorily the adsorption on apatitic tricalcium phosphate, whereas in the case of octocalcium phosphate it is the Freundlich model that describes this phenomenon. The percentages of decolorization are 90% for apatitic tricalcium phosphate and 75% for phosphate octocalcium apatite. Thermodynamic studies indicated that the dye adsorption process onto both phosphates was physisorption, spontaneous and exothermic in nature. Finally, the use of these two phosphates shows a greater potential for the removal of reactive textile dyes, the main advantages; low-cost and easy to prepare in large scale.

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